Year 1 AS Statistics and Mechanics

Chapter 6

Statistical distributions

Practice Booklet

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6.1 Probability distributions

Practice Book, printed page 39.

1. Write down, with a reason, whether or not each of the following is a discrete random variable.

a. The number, X, which shows uppermost when a fair six-sided dice is rolled.

b. The reaction time, T ms, recorded by a participant in an experiment.

c. The number of hours, H, in a day.

Hint

A discrete random variable can randomly take any of a specified range of discrete possible numerical outcomes.

2. A fair six-sided dice is rolled. The number the dice lands on, X, is recorded.
Write the probability distribution of X as:

a. a table

b. a probability mass function.

Geena says that this distribution is an example of a discrete uniform distribution.

c. Explain what is meant by discrete uniform distribution.

Hint

Write down the probability of each of the outcomes in a table, or in the form $\mathrm{P}(X = x) = k$.

3. A discrete random variable, Y, is modelled using the probability mass function

$$\mathrm{P}(Y = y) = \frac{y}{k}, \ y = 1, 2, 3$$

a. Find the value of k.

b. Write down the probability distribution of Y in table form.

c. Find the probability that:

i. $Y > 1$

ii. $Y > 3$

Hint

The sum of all probabilities should equal 1.

4. A discrete random variable X has the probability distribution shown in the table.

x	1	2	3	4	5	6
$\mathrm{P}(X = x)$	0.1	0.15	0.12	0.2	0.22	0.21

Find the probability that:

a. $X \leq 2$

b. $X > 3$

c. $2 \leq X < 6$

Hint

Work out which of the outcomes are included in each inequality.

5. E The random variable, X, has probability function

$$\mathrm{P}(X = x) = kx, \ x = 1, 2, 3, 4, 5$$

a. Show that $k = \frac{1}{15}$ (2 marks)

b. Find $\mathrm{P}(X > 2)$. (2 marks)

6. E/P The table shows the probability distribution of a discrete random variable, Y.

y	1	2	3	4
$\mathrm{P}(Y = y)$	$\alpha + \frac{1}{6}$	$\alpha - \frac{1}{8}$	α	2α

a. Find the value of α. (2 marks)

b. Find $\mathrm{P}(1 < Y < 4)$. (2 marks)

7. E/P A random number generator can generate any integer value from 1 to 20 with equal probability. Given that X is the random variable representing the number generated,

a. write down the name of the distribution of X. (1 mark)

b. Find:

i. $\mathrm{P}(X > 15)$

ii. $\mathrm{P}(2 \leq X \leq 17)$

(2 marks)

Two random numbers are generated.

c. Find the probability that they are both less than 10. (3 marks)

8. E/P Mariah has a biased four-sided dice that can land on one of the numbers 1, 2, 3 or 4. The random variable X represents the number the dice lands on after a single roll.

$$\mathrm{P}(X = r) = \mathrm{P}(X = 5 - r) \text{ for } r = 1, 2$$

a. Given that $\mathrm{P}(X = 1) = 0.28$, find, in table form, the complete distribution of X. (2 marks)

The dice is rolled twice and the totals are added together.

b. Find the probability that the total is 7 or more. (3 marks)

9. E/P A fair seven-sided spinner is spun until it lands on green (G) or it has been spun five times in total.



Find the probability distribution of the random variable X, which represents the number of times the spinner is spun. (4 marks)

6.2 The binomial distribution

Practice Book, printed page 40.

1. Write down, with reasons, whether or not each of the following situations can be modelled using a binomial distribution.

a. The number of heads obtained in 30 tosses of a fair coin.

b. The number of rolls of a fair dice that are needed to obtain a 6.

c. The number of wins achieved by a football team in a total of 20 matches played.

d. The number of overfilled bottles in a sample of 25 taken from a machine that should overfill 2% of bottles.

Hint

The conditions for modelling a situation using a binomial distribution are:

- a fixed number of trials
- two possible outcomes
- a fixed probability of success
- the trials are independent.

2. The random variable $X \sim \mathrm{B}(6, 0.2)$. Find:

a. $\mathrm{P}(X = 2)$

b. $\mathrm{P}(X = 4)$

c. $\mathrm{P}(X < 2)$

Hint

Use your calculator or the formula $\mathrm{P}(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

3. The random variable $Y \sim \mathrm{B}\left(12, \frac{3}{5}\right)$. Find:

a. $\mathrm{P}(Y = 6)$

b. $\mathrm{P}(Y = 10)$

c. $\mathrm{P}(Y \geq 9)$

Hint

$\mathrm{P}(Y \geq 9) = \mathrm{P}(Y = 9) + \mathrm{P}(Y = 10) + \mathrm{P}(Y = 11) + \mathrm{P}(Y = 12)$

4. E Anton spins a fair five-sided spinner 8 times and records the random variable X, the number of times the spinner lands on yellow (Y).



He models this random variable as $X \sim \mathrm{B}(n, p)$.

a. State the values of n and p. (2 marks)

b. Find:

i. $\mathrm{P}(X = 5)$

ii. $\mathrm{P}(X = 3)$

iii. $\mathrm{P}(X \geq 7)$

(3 marks)

5. E/P Charlotte is testing the lifetime of some batteries. The probability that a battery chosen at random lasts less than 30 hours is 0.12. Charlotte takes a sample of 20 batteries.

a. State any assumptions that are required to model the number of batteries that last less than 30 hours as a binomial distribution. (2 marks)

Using this model, find:

b. the probability that exactly 3 batteries last less than 30 hours (2 marks)

c. the probability that two or fewer batteries last less than 30 hours. (3 marks)

6. E/P In a market research poll, it is found that the probability that a person chosen at random likes chocolate spread is 0.8.

A random sample of 50 people is taken from the poll.

a. State a suitable distribution to model the random variable, X, the number of people in the sample who like chocolate spread. (1 mark)

b. Find the probability that:

i. 40 people like chocolate spread

ii. more than 47 people like chocolate spread.

(3 marks)

7. E/P Dawid rolls a fair six-sided dice 10 times. He records, X, the number of times the dice lands on a prime number.

a. Define a suitable distribution to model X and write down two assumptions that are required for the distribution to be valid. (2 marks)

b. Find the probability that Dawid rolls 4 prime numbers. (2 marks)

He repeats the experiment 7 times.

c. Find the probability that he rolls 4 prime numbers exactly twice. (3 marks)

6.3 Cumulative probabilities

Practice Book, printed page 42.

1. The random variable $X \sim \mathrm{B}(12, 0.3)$. Find:

a. $\mathrm{P}(X \leq 2)$

b. $\mathrm{P}(X < 6)$

c. $\mathrm{P}(X \geq 7)$

Hint

Use statistical tables or the binomial cumulative distribution function on your calculator.

2. The random variable $Y \sim \mathrm{B}(15, 0.4)$. Find:

a. $\mathrm{P}(Y \leq 7)$

b. $\mathrm{P}(Y > 9)$

c. $\mathrm{P}(1 < Y \leq 6)$

Hint

$\mathrm{P}(1 < Y \leq 6) = \mathrm{P}(Y \leq 6) - \mathrm{P}(Y \leq 1)$

3. The random variable $T \sim \mathrm{B}(35, 0.61)$. Find:

a. $\mathrm{P}(T < 25)$

b. $\mathrm{P}(T \geq 28)$

c. $\mathrm{P}(15 \leq T < 30)$

Hint

This value of p is not given in the tables so use the binomial cumulative distribution function on your calculator.

4. E The diagram shows a fair spinner.



The spinner is spun 30 times.
Find the probability of it landing on red (R):

a. fewer than 10 times (2 marks)

b. at least 14 times (2 marks)

c. between 10 and 14 times inclusive. (2 marks)

5. E/P Olivia grows orchids. The probability of a particular plant producing fruit is 0.46. Olivia has many greenhouses, each of which holds 25 plants.
A greenhouse is selected at random. Find the probability that in this greenhouse:

a. there are exactly 8 plants with fruit on them (1 mark)

b. more than 12 plants have fruit on them. (2 marks)

Olivia takes a sample of 6 greenhouses.

c. Find the probability that at least 4 of these greenhouses contain more than 12 plants with fruit on them. (3 marks)

6. E/P Annie is calculating the probability of getting at least 4 sixes when she rolls a fair six-sided dice 25 times.
Her working is shown below.

Let X be the random variable 'number of sixes'
$X \sim \mathrm{B}\left(25, \frac{1}{6}\right)$

$$\begin{aligned} \mathrm{P}(X > 4) &= 1 - \mathrm{P}(X \leq 4) \\ &= 1 - 0.5937 \\ &= 0.4063 \end{aligned}$$

a. Identify the mistake that Annie has made. (1 mark)

b. Calculate the correct probability. (2 marks)

7. E/P Pietro rolls a biased six-sided dice 40 times. The probability of rolling a six is 0.2.

a. State a suitable distribution for the random variable X, the number of sixes that he rolls. (1 mark)

b. Find the largest value of k such that $\mathrm{P}(X \leq k) < 0.09$. (2 marks)

c. Find the smallest value of m such that $\mathrm{P}(X > m) < 0.05$. (2 marks)

d. Find $\mathrm{P}(k \leq X \leq m)$. (2 marks)

8. E/P Frankie is playing a computer game. The probability that she collects the hidden box on each level is 0.4.

a. Given that Frankie plays 15 levels, find the probability that:

i. she collects exactly 8 hidden boxes

ii. more than 10 hidden boxes.

(3 marks)

Frankie says the probability that she collects at least 8 boxes is 0.0950.

b. Explain the error that Frankie has made and find the correct probability. (2 marks)

Problem solving: Set A

Bronze

A discrete random variable X has probability mass function $\mathrm{P}(X = x) = \frac{x}{k}, \quad x = 1, 2, 3, 4$

a. Find the value of k. (2 marks)

b. Show the probability distribution of X in table form. (2 marks)

c. Find $\mathrm{P}(X > 2)$. (2 marks)

Silver

A biased four-sided spinner is numbered from 1 to 4. The probability distribution of the random variable X, the outcome when the spinner is spun, is shown in the table.

x	1	2	3	4
$\mathrm{P}(X = x)$	0.2	$2a$	0.35	a

The spinner is spun twice.
Calculate the probability that the total score is:

a. 4 (3 marks)

b. greater than 5. (3 marks)

Gold

In an experiment a group of cats each repeatedly try to catch a toy mouse.
For each cat, the random variable X represents the number of times the cat catches the mouse in the first five attempts. Karen models X as $\mathrm{B}(5, 0.2)$.

a. State two assumptions Karen needs to make to use her model. (2 marks)

b. Using Karen's model, find $\mathrm{P}(X \geq 3)$. (1 mark)

For each cat, the random variable Y represents the number of the attempt on which it first catches the mouse.

c. Using Karen's assumptions about this experiment, find $\mathrm{P}(Y = 3)$. (2 marks)

Ian assumes that in this experiment no cat will need more than 5 attempts to catch the mouse for the first time. He models $\mathrm{P}(Y = n)$ as $\mathrm{P}(Y = n) = 0.1 + (n - 1)k$

d. Find the value of k. (3 marks)

e. Using Ian's model, find $\mathrm{P}(Y = 3)$. (1 mark)

f. Explain how Karen's and Ian's models differ in describing the probability that a cat catches a mouse in this experiment. (1 mark)

Problem solving: Set B

Bronze

A biased coin is tossed 35 times.

a. Given that $\mathrm{P}(\text{heads}) = 0.64$, write down a suitable distribution for the random variable Y, the number of times the coin lands on heads. (1 mark)

b. Find the probability that:

i. $\mathrm{P}(Y \leq 17)$

ii. $\mathrm{P}(5 \leq Y \leq 20)$

(3 marks)

Silver

The random variable $X \sim \mathrm{B}(30, 0.15)$. Find:

a. the largest value of k such that $\mathrm{P}(X < k) < 0.05$ (1 mark)

b. the smallest number r such that $\mathrm{P}(X > r) < 0.01$ (2 marks)

c. $\mathrm{P}(k \leq X \leq r)$. (2 marks)

Gold

A clay pigeon shooter hits a clay with probability 0.65. There are 25 clays in each set. To win a set, the shooter must hit at least 15 clays.
The shooter takes part in 10 sets of 25 clays. Find the probability that the shooter wins:

a. exactly 7 of these sets (3 marks)

b. fewer than 5 of these sets. (2 marks)

Now try this → Exam question bank Q2, Q9, Q14, Q16, Q18, Q28, Q33

Selected exam question bank questions

Question 2

A fair eight-sided dice is rolled 10 times.

a. Write down a suitable distribution for the random variable X, the number of times the dice shows an 8. (1 mark)

b. Calculate:

i. $\mathrm{P}(X = 1)$

ii. $\mathrm{P}(X \geq 2)$

(3 marks)

Question 9

In the large data set, daily mean cloud cover is measured in oktas.

a. Explain what is meant by an 'okta'. (1 mark)

Harriet believes that the daily mean cloud cover in Hurn in 2015 can be modelled by the random variable X with probability mass function

$$\mathrm{P}(X = x) = \frac{1}{8}, \quad x = 1, 2, 3, 4, 5, 6, 7, 8$$

b. State the type of distribution Harriet is using to model X. (1 mark)

c. Use Harriet's distribution to find $\mathrm{P}(X > 6)$. (1 mark)

d. Using your knowledge of the large data set, suggest a reason why Harriet's model might not be suitable. (1 mark)

Question 14

Ishika is taking part in a maths challenge which involves answering a series of multiple choice questions. Each question has four possible answers. Given that there are 25 questions in the challenge and Ishika guesses the answer to each one,

a. define a suitable distribution to model X, the number of questions that Ishika answers correctly (1 mark)

b. find $\mathrm{P}(X \geq 10)$. (2 marks)

A teacher wants to set a pass mark such that the probability of a student passing the test by simply guessing answers is less than 5%.

c. State the lowest pass mark the teacher can set. (2 marks)

Question 16

The discrete random variable X has probability mass function

$$\mathrm{P}(X = x) = \frac{k}{x}, \quad x = 1, 2, 3$$

a. Find the value of k and write down, in table form, the probability distribution of X. (3 marks)

Two independent observations are taken from X.

b. Find the probability that the second observation is larger than the first observation. (2 marks)

Question 18

Egg boxes each contain 24 eggs. The probability that an egg chosen from a box at random is cracked is 0.1.

a. Define a suitable distribution to model the number of eggs, X, that are cracked in a particular box and state two assumptions that are necessary for this distribution to be valid. (2 marks)

A box is selected at random.

b. Find the probability that in this box there are 2 or more cracked eggs. (2 marks)

A sample of 15 boxes is taken.

c. Find the probability that there are exactly 9 boxes with 2 or more cracked eggs in. (2 marks)

Question 28

A bag contains a large number of identical-sized coloured disks.

In the bag, 58% of the disks are blue, 22% of the disks are green and the rest are yellow.

A random sample of 30 disks is taken from the bag.

a. Explain why it is appropriate to model the random variable 'number of yellow disks in the sample' using a binomial distribution. (2 marks)

b. Find the probability that the sample contains at least 7 yellow disks. (2 marks)

All of the disks are replaced and the experiment is repeated.

c. Find the probability that there are at least 7 yellow disks on one occasion and fewer than 7 on the other. (3 marks)

Question 33

A sample of students are asked how many pets they have and the following probability distribution is generated, where X is the discrete random variable 'number of pets'.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	a	0.15	$2a$	0.2	0.1

a. Calculate the value of a. (2 marks)

b. Find the minimum number of students that were surveyed. Give a reason for your answer. (2 marks)

c. Two students are chosen at random from the sample. Find the probability that:

i. both students have 2 pets

ii. between the two students, the total number of pets is 5.

(4 marks)

Answers

Student answers may differ slightly from those shown here depending on whether tables or calculators are used.

6.1 Probability distributions

1.

a. Yes, 6 discrete outcomes, will occur randomly.

b. No, time is a continuous variable.

c. No, since it is fixed as always being 24.

2.

a.

x	1	2	3	4	5	6
$\mathrm{P}(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

b. $\mathrm{P}(X = x) = \frac{1}{6}$, $x = 1, 2, 3, 4, 5, 6$

c. The range of possible outcomes is a specific set of discrete variables where the probability of each occurring is the same.

3.

a. 6

b.

y	1	2	3
$\mathrm{P}(Y = y)$	1/6	2/6	3/6

c.

i. 5/6

ii. 0

4.

a. 0.25

b. 0.63

c. 0.69

5.

a. $k + 2k + 3k + 4k + 5k = 1$ therefore $k = \frac{1}{15}$

b. 4/5

6.

a. 23/120

b. 31/120

7.

a. Discrete uniform distribution

b.

i. 1/4

ii. 4/5

c. 81/400

8.

a.

x	1	2	3	4
$\mathrm{P}(X = x)$	0.28	0.22	0.22	0.28

b. 0.2016

9.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	4/7	12/49	36/343	108/2401	81/2401

6.2 The binomial distribution

1.

a. Yes; fixed number of independent trials, two possible outcomes and fixed probability of success.

b. No; not a fixed number of trials.

c. No; not a fixed probability of success.

d. Yes; fixed number of independent trials, two possible outcomes and fixed probability of success.

2.

a. 0.24576

b. 0.01536

c. 0.65536

3.

a. 0.1766

b. 0.0639

c. 0.2253

4.

a. $n = 8$, $p = 0.8$

b.

i. 0.1468

ii. 0.0092

iii. 0.5033

5.

a. The lifetimes are independent of each other, there are a fixed number of trials, the probability that a battery lasts less than 30 hours is constant.

b. 0.2242

c. 0.5631

6.

a. $X \sim \mathrm{B}(50, 0.8)$

b.

i. 0.1398

ii. 0.0013

7.

a. $X \sim \mathrm{B}(10, 0.5)$ The rolls are independent of each other; the probability of rolling a prime number is constant.

b. 0.2051

c. 0.2803

6.3 Cumulative probabilities

1.

a. 0.2528
b. 0.8822
c. 0.0386

2.

a. 0.7869
b. 0.0338
c. 0.6046

3.

a. 0.8632
b. 0.0138
c. 0.9890

4.

a. 0.8034
b. 0.0082
c. 0.1938

5.

a. 0.0612
b. 0.3429
c. 0.1099

6.

a. She has calculated the probability of greater than 4 sixes, not at least. She has not included the probability that $X = 4$.
b. 0.61846

7.

a. $X \sim \mathrm{B}(40, 0.2)$
b. 4
c. 12
d. 0.9283

8.

a.

i. 0.1181
ii. 0.0093

b. She has worked out the probability of greater than 8 and not included the probability of 8 itself. Correct probability = 0.2131

Problem solving: Set A

Bronze

a. 10

b.

x	1	2	3	4
$\mathrm{P}(X = x)$	1/10	1/5	3/10	2/5

c. 7/10

Silver

a. 0.23
b. 0.34

Gold

a. The probability of a cat catching a mouse is constant; the attempts are independent.
b. 0.05792
c. 0.128
d. 0.05
e. 0.2
f. Karen's model assumes that the probability of catching the mouse is constant whereas Ian's model assumes that the probability increases with each attempt.
OR, in Karen's model the cat may not catch the mouse, whereas in Ian's model the cat will definitely catch the mouse in its first five attempts.

Problem solving: Set B

Bronze

a. $Y \sim \mathrm{B}(35, 0.64)$
b.
i. 0.0043
ii. 0.2491

Silver

a. 2
b. 9
c. 0.9423

Gold

a. 0.2332
b. 0.0127

Selected exam question bank answers

Question 2

a. $X \sim \mathrm{B}(10, \frac{1}{8})$
b.
i. 0.3758
ii. 0.3611

Question 9

a. An 'eighth', so oktas represent the number of eighths of the sky that is cloud-covered.
b. Discrete uniform distribution
c. 0.25
d. It does not include the possibility that X could equal 0.

Question 14

a. $X \sim \mathrm{B}(25, 0.25)$

b. 0.0713

c. 11

Question 16

a. $k = \frac{6}{11}$

x	1	2	3
$\mathrm{P}(X = x)$	6/11	6/22	6/33

b. $\frac{36}{121}$

Question 18

a. $X \sim \mathrm{B}(24, 0.1)$ Each egg must be independent; the probability of an egg being cracked must be constant.

b. 0.7075

c. 0.1392

Question 28

a. There are two outcomes: yellow and not yellow; the probability of success is assumed to be constant and the trials independent since there are a 'large number' of disks and there are a fixed number of trials.

b. 0.3930

c. 0.4771

Question 33

a. $\frac{11}{60}$

b. 60; in lowest terms has a denominator of 60 so since this is discrete data and there cannot be 'part students', 60 is the minimum number of students.

c.

i. $\frac{9}{400}$

ii. $\frac{11}{60}$